Coding Assignment 4

Due Date: *Monday, November 2 (11.59PM)*

Submission: on Gradescope

Instructions

The coding assignments may be completed in groups of *up to 4 students*. If you do so, please make sure that you include everyone's full name, and that you also *select everyone's name when submitting the assignment on Gradescope*. This ensures that each group member will get a grade assigned and have access to the comments from the graders.

For this assignment, the following items need to be submitted:

1. A **single PDF** that contains your code, all results, plots, and written answers (generated from R Markdown, Jupyter, etc.).
2. The **source file** used to produce the PDF (.Rmd, ipynb, ...).

In this problem we are going to work with the Italian olive oil fatty-acid data of Forina, Armanino, Lanteri, and Tiscornia (1983).[1] The data contain $n = 572$ authentic olive oils collected in Italy. Each oil is described by the percentage composition of eight fatty acids in the lipid fraction (`palmitic`, `palmitoleic`, `stearic`, `oleic`, `linoleic`, `linolenic`, `arachidic`, and `eicosenoic`). Oils come from nine growing areas that nest in three broader regions:

Region	Growing areas	n
South	North-Apulia, South-Apulia, Calabria, Sicily	323
Sardinia	Inland-Sardinia, Coastal-Sardinia	98
North	Umbria, East-Liguria, West-Liguria	151

The scientific question is whether fatty-acid composition can recover the geographic origin of an oil. That is a classic authenticity / chemometrics problem, and a natural setting in which to compare Gaussian discriminant rules with a multinomial logistic model.

[1] M. Forina, C. Armanino, S. Lanteri, and E. Tiscornia (1983). Classification of olive oils from their fatty acid composition. In H. Martens and H. Russwurm Jr. (eds.), *Food Research and Data Analysis*, Applied Science Publishers, pp. 189–214.

The file `olive_oils.csv` can be downloaded from Coursera. Use the eight fatty-acid measurements as predictors. The response is the *three-level factor* `Region` (South, Sardinia, North). **Do not** use `Area` as a predictor or as the response.

Two notes on the measurements:

- The fatty-acid columns are stored as percentages $\times 100$ (so they sum to about 10,000 rather than 100). Treat them as eight continuous predictors; no further rescaling is required unless a method you use needs it.
- The eight acids are compositional (they describe parts of a whole). For this assignment you do not need a compositional transformation. Do not interpret a coefficient as if that acid could change while the others stayed fixed.

The **goal** is to compare Discriminant Analysis methods (*linear* and *quadratic*) vs. a Logistic/Multinomial approach to classify the *three* region.

For the questions below, split the data into training and testing (70%–30%) using 598 as a seed. Stratify the split by `Region` so that all three classes appear in both sets. Use the default training-set class priors.

(a) Using the training data fit *three* models: **LDA**, **QDA** and **Multinomial** (Logistic) **regression**. *Note that here the response has three levels, so a binomial logistic is not applicable.*

(b) Report the accuracy for all methods on both training and testing data sets, and prepare confusion matrices for the test predictions. Label the rows and columns by region, not by integer codes. Rows are the true region; columns are the predicted region.

(c) Produce one scatterplot of two fatty acids with points colored by `Region`. Put `eicosenoic` on one axis (the other axis can be oleic or linoleic). Briefly state what the plot suggests about class separation *before* you interpret the fitted models.

(d) Comment on the results. Address all of the following:

- Is there a model that seems to perform better overall on the test set?
- Which model misclassified more observations?
- Which region is classified most / least accurately, and which pair of regions is most often confused?
- Does QDA improve on LDA here, or does the extra covariance structure look like overfit? Relate this to the class sizes and to part (c).

The Sardinia test slice is small (on the order of 30 oils). Comment on the direction of the errors, not on a one- or two-oil swing in accuracy.

Remark: The results might be different between Python and R. That is expected and ok.